

World Happiness Report Dataset

Introduction

This project analyses the World Happiness Report dataset, which provides a structured view of wellbeing through variables such as income, social support, freedom, and generosity. The dataset is widely regarded as a credible source for measuring life satisfaction across countries. The objective of this analysis is to examine how these factors relate to happiness, with a particular focus on identifying patterns among countries that consistently improve in happiness over time.

Limitations

There are several limitations which may make WHR unrepresentative of happiness across the world. As of 2026, there are a total of 195 recognised countries in the world, and the WHR has only included 147 of those countries in the latest findings (in 2025), excluding small or inaccessible populations. Due to this, specific states will be completely unaccounted, possibly skewing the rankings in favour of states with higher populations. Another limitation of the WHR is the subjectivity within the measurements themselves. Although the methods used to gather results for "Life evaluation (3-year average)" are universal, states have different collective perspectives, so ratings may vary. Despite these limitations, the WHR still seems like an accurate representation of worldwide happiness, as each country has some underlying reason for their rank. For instance, Ukraine is still seeing a sharp decline in happiness since their conflict with Russia which started in 2022.

Assumptions

There were a few assumptions that I have made about the dataset.

- The "Life evaluation (3-year average)" is representative of happiness. For any analyses of the WHR dataset, this would have to be valid.
- Multicollinearity was not explicitly addressed. While some correlation between independent variables is expected, its impact on the interpretation of generosity was considered minimal for this analysis.
- 5 entries across a span of 5 years are enough to filter out anomalous observations. After seeing a few outliers with only 2 to 4 entries, I decided to remove them. This comes at a cost of selection bias, but I think it is beneficial as the anomalies would be ranked unfairly due to the other countries having at least 20+ entries.

Cleaning

The cleaning process was focused on preserving the integrity of the dataset while improving usability. This includes excluding confidence intervals to prioritise direct comparison between variables, and standardising feature names as well as removing redundant columns.

Results

I answered three questions during my analysis:

1. Does money (GDP) correlate with happiness? With high-income countries, is the relationship linear or diminishing?
2. Does generosity correlate with happiness? Does it remain significant when accounting for other social, economic factors?
3. What common characteristics define countries that consistently improve in happiness rankings over time, and what factors might explain these trends?

The results prove to be interesting.

Log (GDP) had a moderate positive correlation with happiness (0.682), suggesting that wealth economic freedom is an important but not exclusive driver of life satisfaction. As the population has more access to goods and services, they can satisfy their needs efficiently.

Generosity appears to have no correlation with happiness, removing generosity resulted in a small decrease in model performance by -0.019, it also yielded the lowest coefficient when standardising for all features, which both point to generosity being statistically insignificant in context to happiness. This suggests that individual altruism may have less influence on international happiness.

Social support (having someone to talk to) seems to be correlated with top improvers in happiness, freedom to make life choices sees small increases in improvers, and the other factors seem to be uncorrelated. Notably, GDP isn't a key characteristic in improvers. This highlights the impact that human connection can make on perceived mood.

Overall, this analysis has been intriguing. Discovering that social support being correlated with happier countries, yet generosity being unrelated was surprising. This analysis has heavy implications on what leaders should be focusing on to balance their nationwide satisfaction. And working through this dataset has given me a wider perspective when looking at happiness in different countries, especially in higher-income countries.